



Beachwatch Program quality assurance report

State of the Beaches 2026

Department of Climate Change,
Energy, the Environment and Water



Acknowledgement of Country

Department of Climate Change, Energy, the Environment and Water acknowledges the Traditional Custodians of the lands where we work and live.

We pay our respects to Elders past, present and emerging.

This resource may contain images or names of deceased persons in photographs or historical content.

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Artist and designer Nikita Ridgeway from Aboriginal design agency Boss Lady Creative Designs created the People and Community symbol.

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Contents

Introduction	1
The quality assurance program	2
Field sampling	2
Laboratory analysis	5
Data management	8
Community reporting	9
References	13
More information	14

List of figures

Figure 1	Percentage compliance with Beachwatch sampling protocols in 2025–26	4
Figure 2	Distribution of duplicate enterococci (cfu/100 mL) results for the contracted laboratory, May 2025 to April 2026	7
Figure 3	Distribution of duplicate enterococci (cfu/100 mL) results for the contracted laboratory of Kiama Municipal Council and Shoalhaven City Council, May 2025 to February 2026	8
Figure 4	Percentage compliance with beach pollution forecast reporting protocols in 2025–26	11

Introduction

Recreational water quality in New South Wales has been monitored by the Department of Climate Change, Energy, the Environment and Water's Beachwatch Program since 1989. Since 2002, the program has operated in partnership with local councils, government agencies, and wastewater managers through the Beachwatch Partnership Program.

This report summarises the outcomes of the Beachwatch quality assurance program for the 2025–26 period, covering field sampling, laboratory analysis, data management, and community reporting.

The purpose of the quality assurance program is to ensure the accuracy, consistency and reliability of water quality data and reporting. This enables the public to make informed decisions about when and where to swim and supports ongoing confidence in the Beachwatch Program.

Overall, quality assurance was maintained to a high standard throughout the reporting year.



Image 1 **Water sample collection by the Beachwatch Program. Photo: Beachwatch/DCCEEW**

The quality assurance program

To ensure the accuracy and reliability of data reported by Beachwatch, quality assurance is integrated across all components of the program, including:

- field sampling (equipment preparation, sample collection, storage and transport)
- laboratory analysis
- data management
- community reporting.

Beachwatch reporting system

The Beachwatch reporting system streamlines, integrates and automates data capture to support real-time reporting of water quality information. The system consists of multiple applications, including a website, a Salesforce platform with a partner portal, MuleSoft, and a database, all with stringent quality controls built in, which enhance quality assurance for the program.

Field sampling

Sampling audits

Hunter Water, Sydney Water and Beachwatch collect samples year-round and are audited quarterly. Councils participating in the Beachwatch Partnership Program typically conduct sampling during all or part of the swimming season (October to April) and are audited once during this period.

Audits assess field officer performance against established Beachwatch sampling protocols. These include aseptic sampling techniques, sample collection and storage, and documentation of field observations.

The protocols are based on internationally recognised methods for collecting water samples in recreational bathing areas (APHA, 1998).

Beachwatch

During the 2025–26 sampling season, 4 field audits were completed on Beachwatch field officers. Sample collection by Beachwatch field officers complied well with established sampling protocols, with a compliance of 100% (Figure 1). This is an excellent result and shows samplers have a high level of competency when field sampling.

Beachwatch partners

Eight field audits were completed during the 2025–26 sampling season on wastewater managers, Hunter Water and Sydney Water, sampling officers. Sample collection fully complied with established sampling protocols, achieving 100% compliance (Figure 1).

During the 2025–26 sampling season, 11 field audits were completed on councils in the Beachwatch Partnership Program, achieving an overall compliance of 96% with Beachwatch sampling protocols (Figure 1).

During 6 council audits, it was observed that some samples may not have been collected using the correct aseptic techniques or stored at the appropriate temperature.

In response, samplers were reminded of the correct aseptic sampling procedures and demonstrated appropriate sample collection techniques during the field audit. Partners were also advised to monitor the esky temperatures to ensure samples remained within the required temperature range during storage and transport.

Who samples where?

Beachwatch

Collects samples at 97 ocean and harbour beaches in Sydney.

Hunter Water

Collects samples at 17 ocean beaches in Port Stephens, Newcastle and Lake Macquarie.

Sydney Water

Collects samples at 18 ocean beaches in Wollongong, Shellharbour and Kiama.

Partner councils

Ballina Shire Council, Bathurst Regional Council, Blue Mountains Council, Central Coast Council, Clarence Valley Council, Eurobodalla Shire Council, Hawkesbury City Council, Kiama Municipal Council, Parkes Shire Council, Richmond Valley Council, Shoalhaven City Council and Wollongong City Council collect samples at 111 popular swimming locations in their respective local government areas.

State government partners

Open Spaces Program and Placemaking NSW of the Department of Planning, Housing and Infrastructure funded sample collection at 2 swim sites.

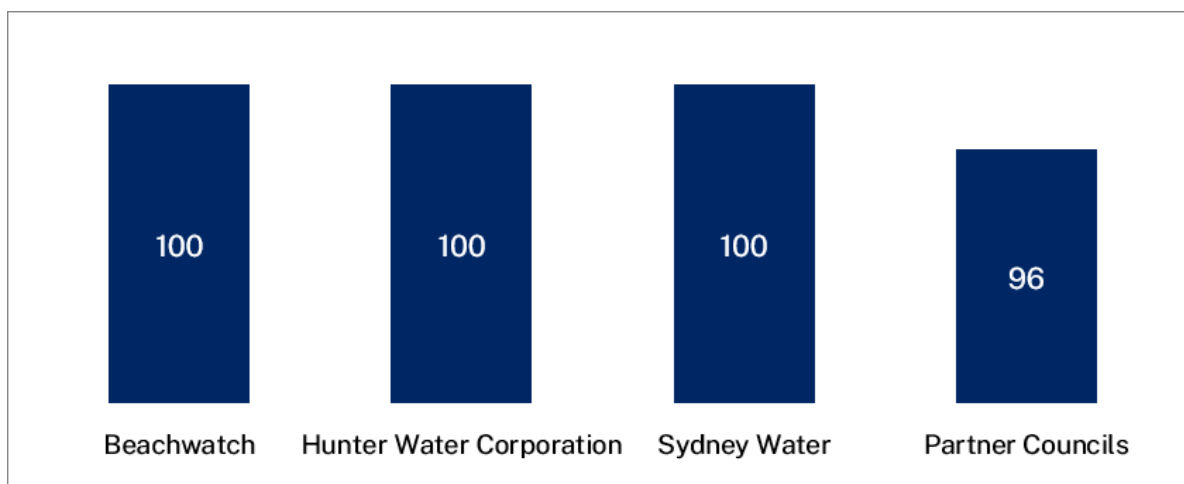


Figure 1 Percentage compliance with Beachwatch sampling protocols in 2025–26

Quality control samples

Quality control samples such as field blanks and trip blanks are recommended in water quality monitoring programs to detect sample contamination during collection, storage and transportation of samples. This is in line with established Beachwatch quality assurance protocols based on internationally recognised methods for collecting water samples in recreational bathing areas (APHA, 1998).

Trip blanks are containers filled with sterilised water used to assess cross-contamination during storage and transport of water samples during the sampling run. Field blanks are collected during the sample run by the sampler by filling a sample bottle with sterilised water to detect for contamination by the sampler.

Beachwatch Program

In the 2025–26 sampling season, quality control samples (field blanks and trip blanks) were collected by Beachwatch on 358 occasions to assess potential contamination during sample collection, storage and transport. Only one of these 358 quality control samples indicated possible cross-contamination. As a precaution, the enterococci results for the 7 samples collected during that sampling run were excluded from reporting. This is an excellent result and demonstrates the consistently high standard of sampling practices maintained by Beachwatch field officers.

Beachwatch partners

In the 2025–26 sampling season, 300 quality control samples (field blanks and trip blanks) were collected by partners. No contamination was detected in any quality control samples collected. This is an excellent result, and confidence can be placed in the sampling activities of partners.

Laboratory analysis

Laboratory proficiency testing

Laboratories used by Beachwatch and our partners are required to undertake proficiency testing to ensure confidence in the data reported. To ensure test results for water quality sampling are reliable, we recommend using laboratories accredited by the National Association of Testing Authorities (NATA). As part of this accreditation, the laboratory participates in regular proficiency testing to demonstrate the accuracy and consistency of its results.

For partners using non-accredited laboratories, it is important that these laboratories undertake independent proficiency testing to ensure the accuracy and reliability of their results.

Beachwatch Program

All Beachwatch water samples are analysed by a commercial laboratory accredited by the National Association of Testing Authorities (NATA). As part of this accreditation, the laboratory participates in regular proficiency testing to demonstrate the accuracy and consistency of its results and maintained its accreditation.

Beachwatch partners

In 2025–26, partner laboratories were either NATA accredited or participated in regular proficiency testing programs to demonstrate competence in enterococci analysis.

Water samples from the following Beachwatch partners were analysed by NATA accredited laboratories that comply with strict quality assurance requirements:

- Ballina Shire Council (partial season)
- Blue Mountains Council
- Clarence Valley Council (partial season)
- Richmond Valley Council (partial season)
- Hawkesbury City Council
- Wollongong City Council
- Kiama Municipal Council
- Shoalhaven City Council
- Eurobodalla Shire Council
- Parkes Shire Council
- Bathurst Regional Council
- Hunter Water
- Sydney Water
- Open Spaces Program, Department of Planning, Housing and Infrastructure
- Placemaking NSW, Department of Planning, Housing and Infrastructure.

The proficiency testing for Central Coast Council's 2 laboratories was conducted by IFM Quality Services Pty Ltd. IFM Quality Services despatches samples in freeze-dried form that require reconstitution before testing.

The councils' laboratories underwent regular proficiency testing during May 2025 to April 2026 and recorded good results with enterococci counts reported within defined limits. Confidence can be placed in the accuracy of data from these laboratories.

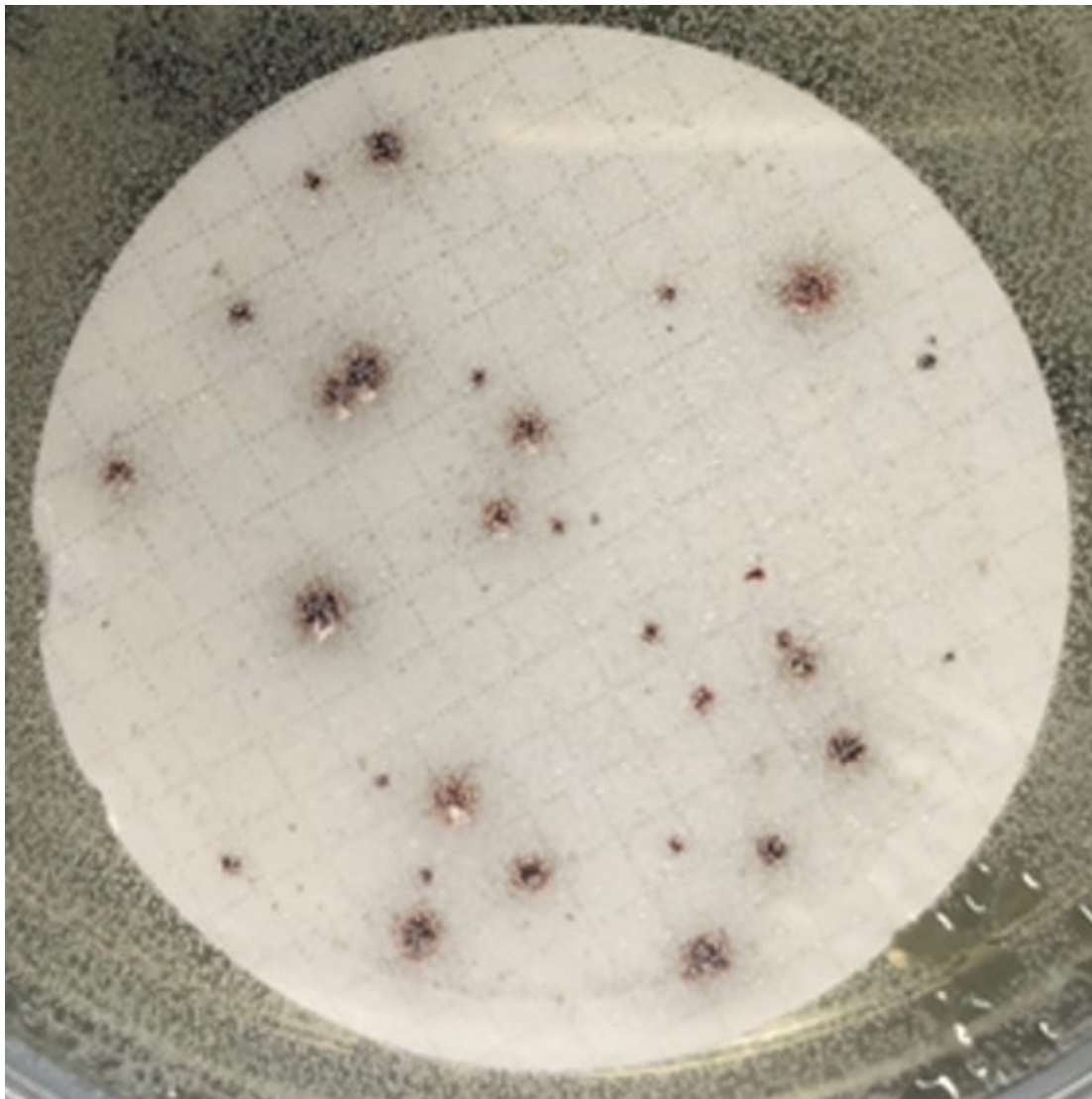


Image 2 Confirmed colonies of enterococci on plate. Photo: Silliker Australia

Duplicate sampling

Beachwatch and partners may independently check the reliability of sample analysis by the laboratory by submitting duplicate water samples to the laboratory. Duplicate samples are collected from the same site at the same time, and the laboratory is not informed that the samples originate from a single location. The results are expected to be similar.

Given the natural variability of bacterial levels in environmental samples, duplicate results within 0.3 log-units of each other (equivalent to a halving or doubling of concentration on a linear scale) are considered acceptable.

Beachwatch Program

Beachwatch submitted duplicate water samples to our contracted microbiological laboratory, which is accredited by the National Association of Testing Authorities (NATA).

During 2025–26, 76% of duplicate samples were within the acceptable limit, which is a good result given the natural variability of bacterial levels. Of the 135 enterococci samples, 24% were outside the acceptable limit. Of those, the majority were at very low bacterial densities and below the safe swimming threshold.

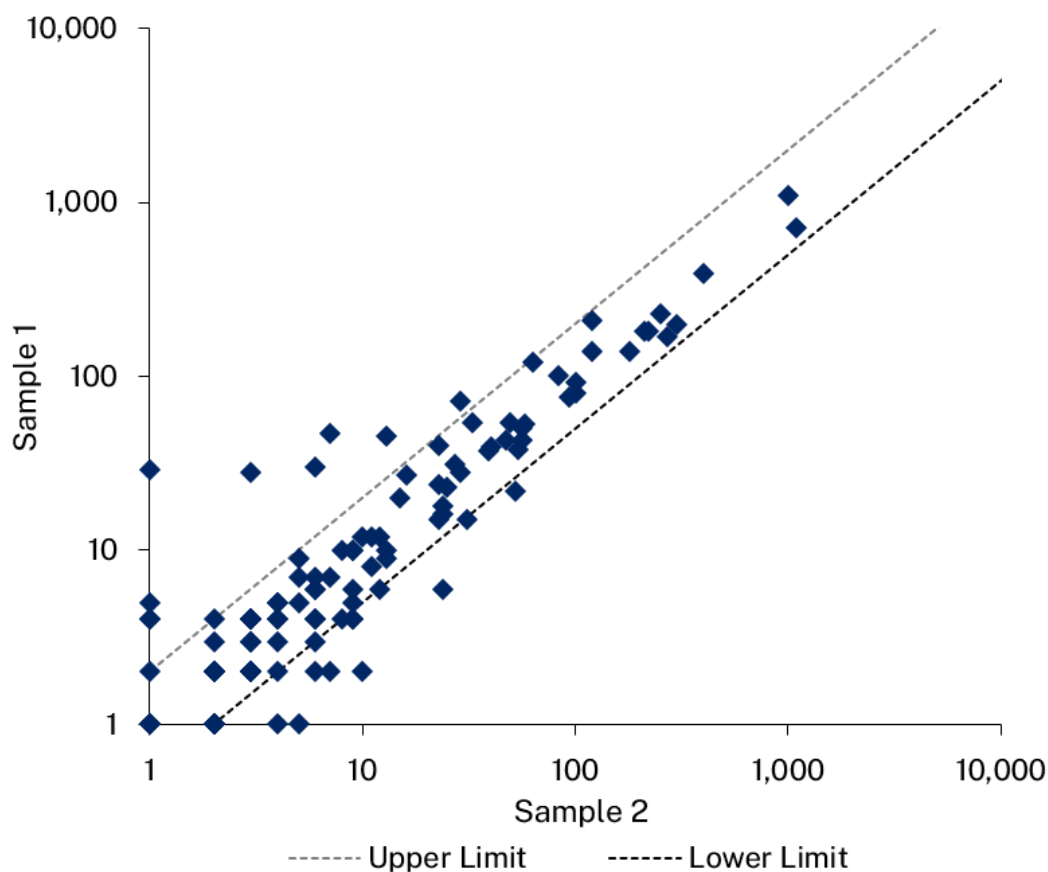


Figure 2 Distribution of duplicate enterococci (cfu/100 mL) results for the contracted laboratory, May 2025 to April 2026

Beachwatch partners

Kiama Municipal Council and Shoalhaven City Council submitted duplicate samples to assess the reliability of laboratory data from their NATA accredited laboratory.

During 2025–26, 85% of the duplicate samples were within the acceptable limit, which is an excellent outcome given the inherent variability of bacteria concentrations in recreational waters. While 15% of the 27 duplicate enterococci sample pairs were outside the acceptable limit, the majority of these occurred at very low bacterial concentrations and remained below the recreational water quality guideline for safe swimming.

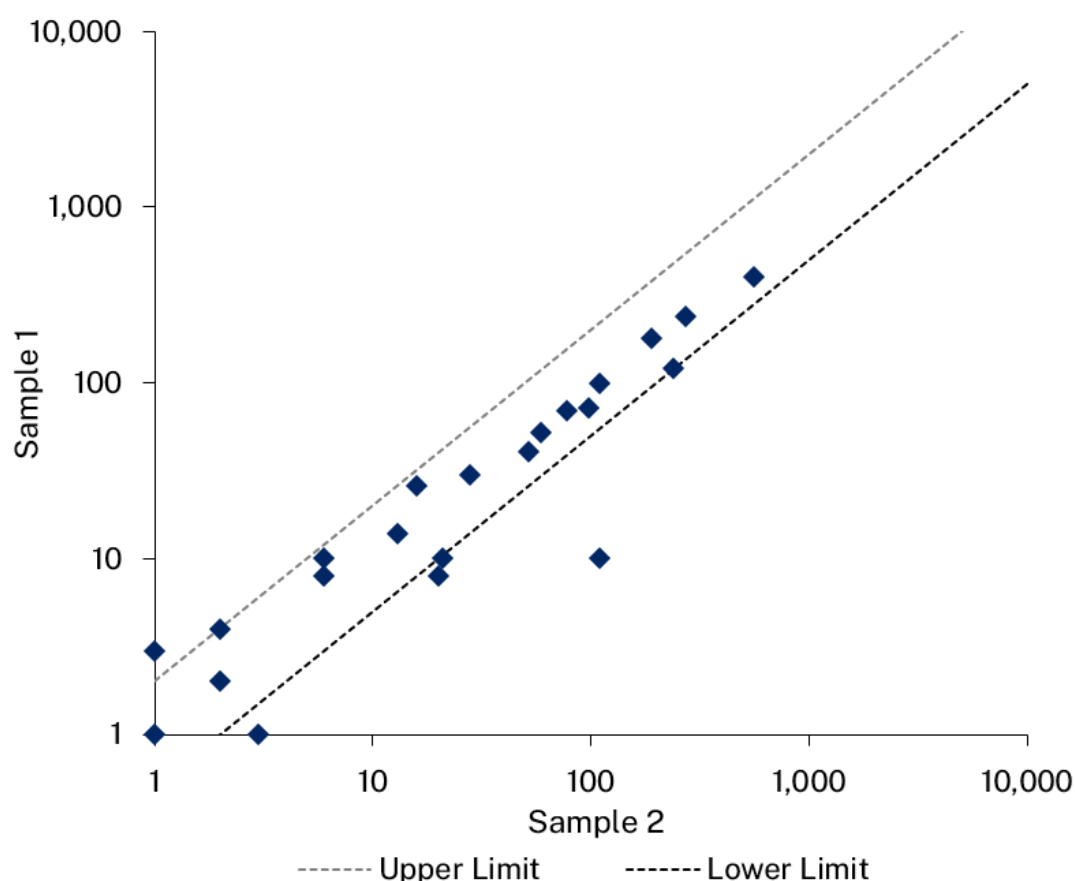


Figure 3 Distribution of duplicate enterococci (cfu/100 mL) results for the contracted laboratory of Kiama Municipal Council and Shoalhaven City Council, May 2025 to February 2026

Data management

Quality assurance procedures for data management within the Beachwatch reporting system ensure that rigorous protocols are followed across all aspects of the program.

Water quality data for monitored swim sites in the Beachwatch Program are entered into the Beachwatch reporting system, built on the Salesforce platform, including the partner portal, by both Beachwatch and its partners.

For Illawarra beaches, Sydney Water electronically submits data directly from its laboratory to the Beachwatch team, who then upload it to the reporting system.

To minimise manual handling and reduce the risk of data entry errors, Sydney Water uses a data template specifically designed to be compatible with the Beachwatch reporting system.

Several quality controls have been built into the reporting system to strengthen data management processes, including:

- Direct data entry by responsible personnel: field observations and water quality data are entered by the person responsible, reducing the potential for manual data handling errors by other users.
- Systematic quality checks: all data undergoes a series of quality checks, including review by the original user and cross-checking by another user. Once users complete these checks, the system prevents further changes to the data.
- Validation rules and mandatory fields: built-in system requirements ensure data is not duplicated and that all necessary fields are completed using acceptable formats and values.
- User account permissions: access is controlled through user-specific permissions, which restrict who can enter, modify or approve data.
- Template-based uploads: the system can ingest data using fixed templates when required, further minimising manual entry and associated errors.

Once water quality data passes the quality checks and is approved, it is automatically synchronised to the secure Beachwatch database and published on the Beachwatch website. The results are then available for the public to view and download.

Beachwatch conducts monthly data validation checks to confirm that all water quality information has been submitted and uploaded into the Beachwatch reporting system.

Community reporting

Providing the community with up-to-date water quality information is a core function of the Beachwatch Program, and reporting has been integrated into the overall quality assurance framework.

This approach enables Beachwatch to monitor the accuracy and reliability of its service. Where necessary, insights from this process are used to refine and improve reporting practices.

Beachwatch website

The Beachwatch website was launched in August 2023 as part of the integrated Beachwatch reporting system. The system streamlines, integrates, and automates data capture to support real-time reporting of water quality information, including water quality forecasts and the latest star ratings.

There are 2 main types of Beachwatch reporting:

- water quality forecasts
- star ratings reports.

Water quality forecasts

Water quality forecasts provide guidance to help beach users decide when and where it is safe to swim. Forecasts are generated twice daily and assess the likelihood of bacterial contamination at swimming sites across the Hunter, Central Coast, Sydney and Illawarra regions. This information is available to the public via the Beachwatch website and is also shared on X (formerly Twitter), Facebook, and by email to subscribers.

The forecasts are based on telemetered rainfall data and reports of pollution incidents that may affect swim site water quality. They include predictions of the likelihood of pollution at ocean beaches and estuarine swimming areas, along with daily weather, tide and coastal condition forecasts provided by the Australian Bureau of Meteorology.

Forecasts are updated throughout the day when conditions change, using information from the Bureau of Meteorology, local councils, lifeguards, the Environment Protection Authority and Sydney Water.

Accuracy of water quality forecast predictions

Daily water quality forecast scenarios are validated against bacterial monitoring data to assess model performance and track prediction accuracy. In 2025–26, 93% of overall predictions were correct; consistent with results from the previous reporting year (Figure 4).

Forecasts for Pittwater swimming sites were the most accurate, with 96% of scenarios correctly predicted. Illawarra, Hunter and Sydney beaches and Sydney Harbour swimming sites also performed strongly, with between 94-95% accuracy in forecast scenarios.

When the accuracy of pollution predictions for a swimming site declines, the forecasting models are reviewed and adjusted to reflect changes in water quality patterns. Ongoing tracking of prediction accuracy ensures that a high level of overall forecast reliability is maintained.

Reliability of water quality forecasts

Water quality forecasts are generated automatically twice daily and rely on the integration of multiple data sources, including the Bureau of Meteorology, Sydney Water, Manly Hydraulics Laboratory, and applications that support reporting on the Beachwatch website. Forecast subscription emails are sent to subscribers each morning.

The forecasts are monitored daily to ensure any issues are promptly identified and resolved, minimising disruptions. If a forecast cannot be updated for more than 24 hours, the affected swim site will display the message 'Forecast not available' on the Beachwatch website.

- Forecast reliability is measured by the number of times a water quality forecast is successfully produced out of the total number of expected forecasts. This ensures that the information provided remains current and reliable.
- Email subscription reliability is measured by the number of forecast subscription emails sent on time, helping to ensure subscribers receive timely and accurate information.

Between May 2025 and April 2026, 98.66% of the automated morning and afternoon water quality forecasts (116,800 predictions) were successfully generated for 160 swim sites.

Forecasts were unavailable 1.34% of the time (1,568 occasions), primarily due to technical issues such as unavailable rainfall data or connectivity disruptions with data sources. This high level of performance is a strong result, given the complexity of the reporting system, which relies on multiple data sources, applications and automated workflows.

In the same period, 99.5% of forecast subscription emails were sent on time to subscribers. There were 2 instances where daily forecast emails were delayed due to technical issues. A breakdown of forecast performance on the Beachwatch website and via email is provided in Figure 4.

Beachwatch's water quality forecasting service is actively monitored each day by the Beachwatch team. Any issues that impact service delivery are promptly raised, investigated, and addressed in collaboration with relevant support teams and stakeholders. All issues are documented in an electronic tracking system. During 2025–26, following a review, several alternative rainfall gauges from other providers were added as backup at selected Sydney swim sites to improve the reliability and availability of forecasts.

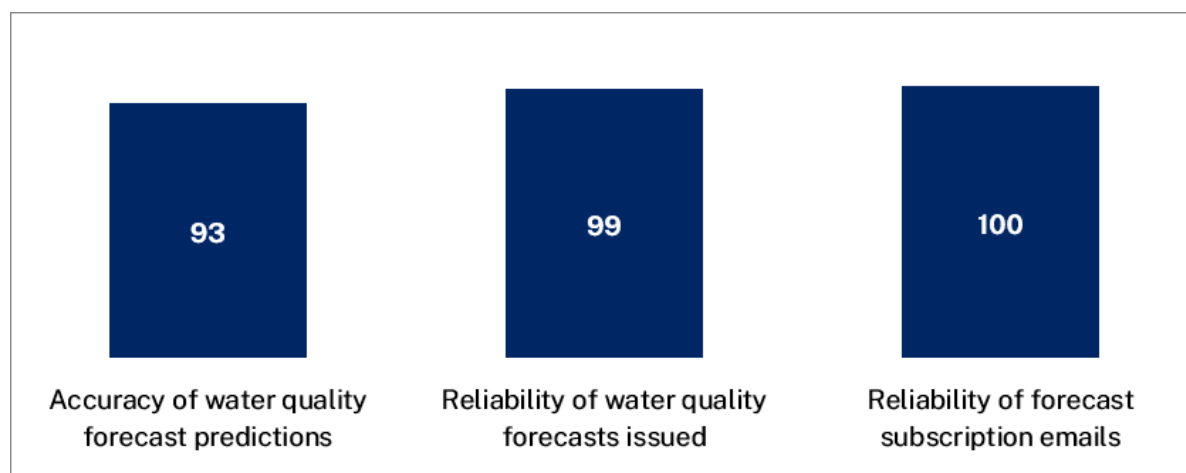


Figure 4 Percentage compliance with beach pollution forecast reporting protocols in 2025–26

Star ratings reports

Star rating reports provide an indication of recent bacterial water quality at swimming sites, based on the National Health and Medical Research Council (NHMRC) 2008 guidelines.

Ratings range from one star, indicating poor water quality, to 4 stars, indicating excellent water quality. The latest rating for each site is based on the enterococci level in the most recent water sample.

Once a water sample passes quality assurance checks and is uploaded to the Beachwatch database, the star rating is automatically published on the Beachwatch website for each monitored site.

Star ratings are updated weekly during the summer swimming season. In the Sydney, Hunter and Illawarra regions, updates generally continue throughout the year. In regional areas and during periods of reduced monitoring, such as winter, ratings may be updated less frequently and only when new water quality results become available. All historical enterococci water quality data is also available on the Beachwatch website.



Image 3 Beach warning signs. Photo: Beachwatch/DCCEEW

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More information

- [About star ratings for beach water quality](#)
- [Beachwatch NSW on X \(formerly Twitter\)](#)
- [Beachwatch NSW on Facebook](#)
- [Beachwatch webpage](#)
- [Water quality data download](#)
- [Sanitary inspection of beaches](#)
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